

Title

Antimicrobial oral rinse formulation demonstrates suppression of cariogenic bacterial colonies in randomized trial

Abstract

Objective: Herbal extract-based oral preparations have traditional antimicrobial properties, yet clinical evidence remains sparse. This parallel-group double-blind investigation compared a herbal rinse formulation against placebo for its capacity to inhibit oral pathogens associated with tooth decay. **Methods:** Sixty participants were randomly allocated to daily oral rinsing with either a botanical extract or isotonic saline for one week. Microbiological sampling and caries-activity testing occurred before and after the intervention period. **Results:** The botanical extract group exhibited significantly reduced bacterial counts compared with the saline-treated control group ($p < 0.05$). Demineralization markers were equivalent between groups, indicating no adverse effects on enamel integrity. The active rinse showed broad antibacterial efficacy against the target pathogens without systemic absorption. **Conclusion:** The herbal formulation demonstrates promising antimicrobial activity in the oral cavity with a favorable safety profile, supporting its investigation as an adjunctive prevention strategy.